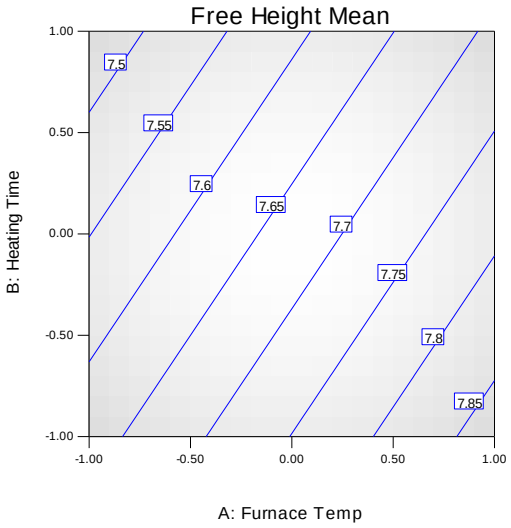
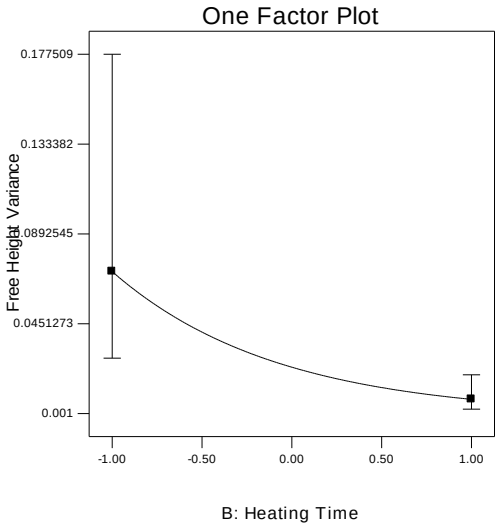
Chapter 12
Robust Parameter Design and Process Robustness Studies Solutions

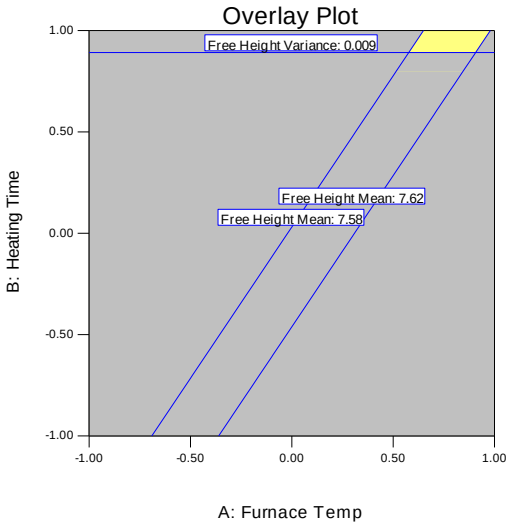
12.1. Reconsider the leaf spring experiment in Table 12.1. Suppose that the objective is to find a set of conditions where the mean free height is as close as possible to 7.6 inches with a variance of free height as small as possible. What conditions would you recommend to achieve these objectives?

A	*B*	*C*	*D*	*E*(–)	*E*(+)	$\bar{y}$	s^2
–	–	–	–	7.78,7.78, 7.81	7.50, 7,25, 7.12	7.54	0.090
+	–	–	+	8.15, 8.18, 7.88	7.88, 7.88, 7.44	7.90	0.071
–	+	–	+	7.50, 7.56, 7.50	7.50, 7.56, 7.50	7.52	0.001
+	+	–	–	7.59, 7.56, 7.75	7.63, 7.75, 7.56	7.64	0.008
–	–	+	+	7.54, 8.00, 7.88	7.32, 7.44, 7.44	7.60	0.074
+	–	+	–	7.69, 8.09, 8.06	7.56, 7.69, 7.62	7.79	0.053
–	+	+	–	7.56, 7.52, 7.44	7.18, 7.18, 7.25	7.36	0.030
+	+	+	+	7.56, 7.81, 7.69	7.81, 7.50, 7.59	7.66	0.017

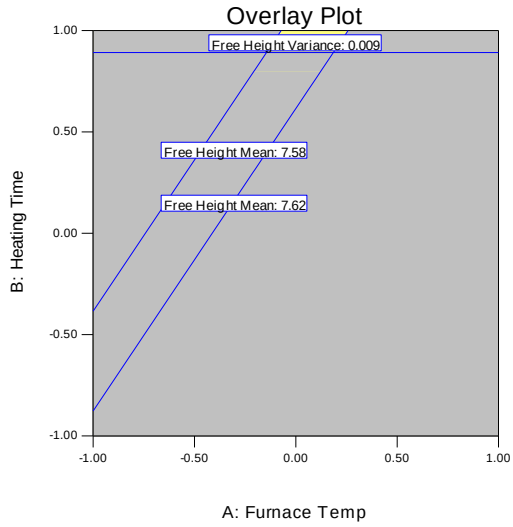
By overlaying the contour plots for Free Height Mean and the Free Height Variance, optimal solutions can be found. To minimize the variance, factor *B* must be at the high level while factors *A* and *D* are adjusted to assure a mean of 7.6. The two overlay plots below set factor *D* at both low and high levels. Therefore, a mean as close as possible to 7.6 with minimum variance of 0.008 can be achieved at *A* = 0.78, *B* = +1, and *D* = -1. This can also be achieved with *A* = +0.07, *B* = +1, and *D* = +1.







Factor D = -1



Factor D = +1

12.2S. Consider the bottle filling experiment in Problem 6.20. Suppose that the percentage of carbonation (A) is a noise variable ($\sigma_z^2 = 1$ in coded units).

(a) Fit the response model to these data. Is there a robust design problem?

The following is the analysis of variance with all terms in the model followed by a reduced model. Because the noise factor A is significant, and the AB interaction is moderately significant, there is a robust design problem.

Design Expert Output

Response: Fill Deviation

ANOVA for Response Surface Reduced Cubic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Cor Total	300.05	3				
Model	73.00	7	10.43	16.69	0.0003	significant
A	*36.00*	*1*	*36.00*	*57.60*	*< 0.0001*	
B	*20.25*	*1*	*20.25*	*32.40*	*0.0005*	
C	*12.25*	*1*	*12.25*	*19.60*	*0.0022*	
AB	*2.25*	*1*	*2.25*	*3.60*	*0.0943*	
AC	*0.25*	*1*	*0.25*	*0.40*	*0.5447*	
BC	*1.00*	*1*	*1.00*	*1.60*	*0.2415*	
ABC	*1.00*	*1*	*1.00*	*1.60*	*0.2415*	
Pure Error	5.00	8	0.63			
Cor Total	78.00	15				

Based on the above analysis, the AC, BC, and ABC interactions are removed from the model and used as error.

Design Expert Output

Response: Fill Deviation

ANOVA for Response Surface Reduced Cubic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	70.75	4	17.69	26.84	< 0.0001	significant
A	*36.00*	*1*	*36.00*	*54.62*	*< 0.0001*	

B	*20.25*	*1*	*20.25*	*30.72*	*0.0002*	
C	*12.25*	*1*	*12.25*	*18.59*	*0.0012*	
AB	*2.25*	*1*	*2.25*	*3.41*	*0.0917*	
Residual	7.25	11	0.66			
Lack of Fit	2.25	3	0.75	1.20	0.3700	not significant
Pure Error	5.00	8	0.63			
Cor Total	78.00	15				

The Model F-value of 26.84 implies there is a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.81	R-Squared	0.9071
Mean	1.00	Adj R-Squared	0.8733
C.V.	81.18	Pred R-Squared	0.8033
PRESS	15.34	Adeq Precision	15.424

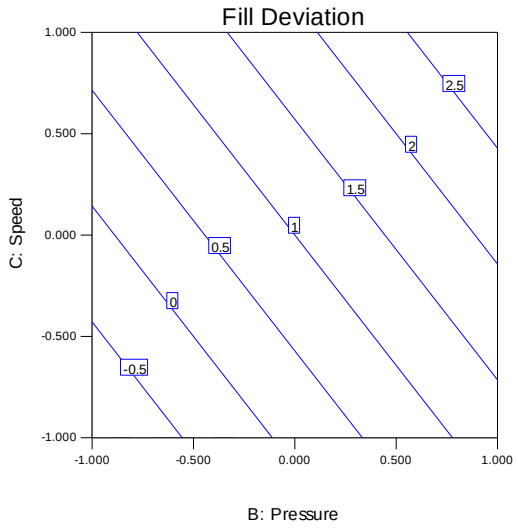
Final Equation in Terms of Coded Factors:

Fill Deviation =
+1.00
+1.50 * A
+1.13 * B
+0.88 * C
+0.38 * A * B

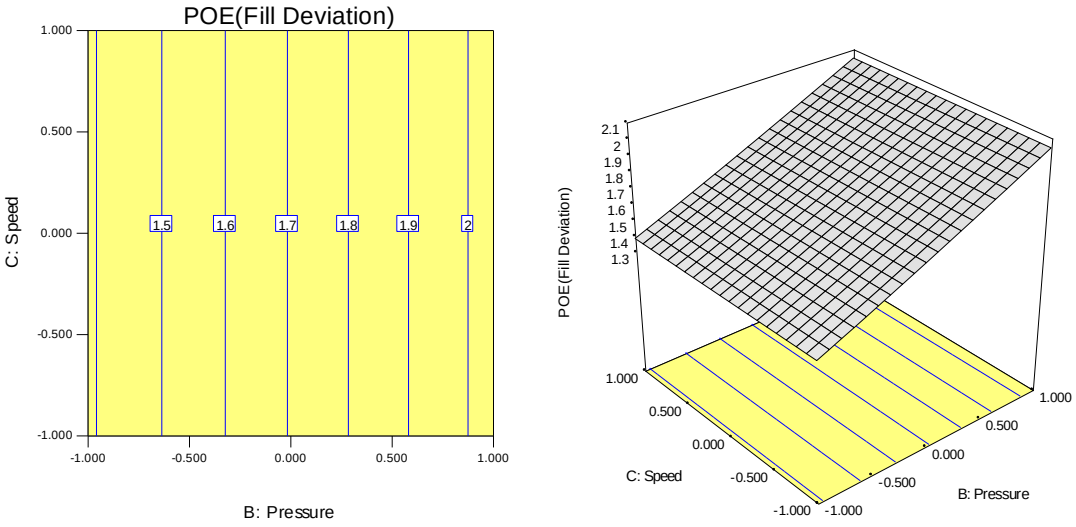
(b) Find the mean model and either the variance model or the POE.

From the final equation shown in the above analysis, the mean model and corresponding contour plot is shown below.

$$E_z\left[y\left(\mathbf{x}, z_1\right)\right] = 1 + 1.13x_2 + 0.88x_3$$

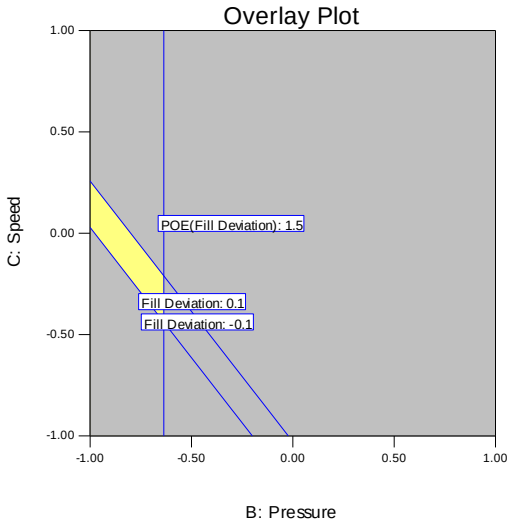


Contour and 3-D plots of the POE are shown below.



(c) Find a set of conditions that result in mean fill deviation as close to zero as possible with minimum transmitted variance.

The overlay plot below identifies a an operating region for pressure and speed that in a mean fill deviation as close to zero as possible with minimum transmitted variance.



12.3. Reconsider the leaf spring experiment from Table 12.1. Suppose that factors *A*, *B* and *C* are controllable variables, and that factors *D* and *E* are noise factors. Set up a crossed array design to investigate this problem, assuming that all of the two-factor interactions involving the controllable variables are thought to be important. What type of design have you obtained?

The following experimental design has a 2^3 inner array for the controllable variables and a 2^2 outer array for the noise factors. A total of 32 runs are required.

	Outer Array				
Inner Array	*D*	-1	1	-1	1

A	*B*	*C*	*E*	-1	-1	1	1
-1	-1	-1					
1	-1	-1					
-1	1	-1					
1	1	-1					
-1	-1	1					
1	-1	1					
-1	1	1					
1	1	1					

12.4. Continuation of Problem 12.3. Reconsider the leaf spring experiment from Table 12.1. Suppose that *A*, *B* and *C* are controllable factors and that factors *D* and *E* are noise factors. Show how a combined array design can be employed to investigate this problem that allows all two-factor interactions to be estimated and only requires 16 runs. Compare this with the crossed array design from Problem 12.3. Can you see how in general combined array designs have fewer runs than crossed array designs?

The following experiment is a 2^{5-1} fractional factorial experiment where the controllable factors are *A*, *B*, and *C* and the noise factors are *D* and *E*. Only 16 runs are required versus the 32 runs required for the crossed array design in Problem 12.3.

A	*B*	*C*	*D*	*E*	Free Height
–	–	–	–	+	
+	–	–	–	–	
–	+	–	–	–	
+	+	–	–	+	
–	–	+	–	–	
+	–	+	–	+	
–	+	+	–	+	
+	+	+	–	–	
–	–	–	+	–	
+	–	–	+	+	
–	+	–	+	+	
+	+	–	+	–	
–	–	+	+	+	
+	–	+	+	–	
–	+	+	+	–	
+	+	+	+	+	

12.5. Consider the connector pull-off force experiment shown in Table 12.2. Show how an experiment can be designed for this problem that will allow a full quadratic model to be fit in the controllable variables along all main effects of the noise variables and their interactions with the controllable variables. How many runs will be required in this design? How does this compare with the design in Table 12.2?

There are several designs that can be employed to achieve the requirements stated above. Below is a small central composite design with the axial points removed for the noise variables. Five center points are also included which brings the total runs to 35. As shown in the alias analysis, the full quadratic model for the controllable variables is achieved.

A	*B*	*C*	*D*	*E*	*F*	*G*
+1	+1	+1	-1	+1	+1	+1
+1	+1	-1	+1	-1	+1	-1
+1	+1	-1	+1	+1	-1	+1

+1	-1	+1	+1	-1	+1	+1
-1	+1	+1	-1	-1	+1	-1
+1	-1	-1	-1	+1	-1	-1
-1	+1	-1	+1	+1	-1	+1
+1	+1	+1	+1	-1	+1	-1
+1	-1	+1	-1	-1	-1	+1
-1	-1	-1	-1	+1	+1	-1
-1	+1	-1	+1	-1	-1	-1
+1	+1	+1	-1	+1	-1	-1
+1	-1	-1	+1	-1	-1	-1
-1	-1	+1	-1	-1	-1	+1
-1	+1	-1	-1	-1	+1	+1
+1	-1	-1	-1	-1	+1	+1
-1	+1	-1	-1	+1	+1	+1
+1	-1	-1	+1	+1	+1	+1
-1	-1	+1	+1	+1	-1	+1
-1	-1	+1	+1	+1	+1	-1
-1	+1	+1	+1	-1	-1	+1
-1	-1	-1	-1	-1	-1	-1
-2.17	0	0	0	0	0	0
2.17	0	0	0	0	0	0
0	-2.17	0	0	0	0	0
0	2.17	0	0	0	0	0
0	0	-2.17	0	0	0	0
0	0	2.17	0	0	0	0
0	0	0	-2.17	0	0	0
0	0	0	2.17	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Design Expert Output

Alias Matrix

[Est. Terms]	**Aliased Terms**
[Intercept]	= Intercept
[A]	= A
[B]	= B
[C]	= C
[D]	= D
[E]	= E + 0.211 * EG + 0.789 * FG
[F]	= F - EF - EG
[G]	= G - EF - 0.158 * EG + 0.158 * FG
[A^2]	= A^2
[B^2]	= B^2
[C^2]	= C^2
[D^2]	= D^2
[E^2]	= $E^2 + F^2 + G^2$
[AB]	= AB - 0.105 * EG - 0.895 * FG
[AC]	= AC - 0.158 * EG + 0.158 * FG
[AD]	= AD + 0.421 * EG + 0.579 * FG
[AE]	= AE - 0.474 * EG + 0.474 * FG
[AF]	= AF + EF + 1.05 * EG - 0.0526 * FG
[AG]	= AG + EF + 1.05 * EG - 0.0526 * FG
[BC]	= BC - 0.263 * EG + 0.263 * FG
[BD]	= BD - EF - 0.158 * EG + 0.158 * FG
[BE]	= BE - 0.368 * EG + 0.368 * FG
[BF]	= BF + 1.11 * EG - 0.105 * FG
[BG]	= BG + EF + 0.421 * EG - 0.421 * FG
[CD]	= CD - 0.421 * EG + 0.421 * FG
[CE]	= CE - EF + 0.158 * EG + 0.842 * FG
[CF]	= CF - EF - 0.211 * EG + 0.211 * FG

[CG] = CG - 1.21 * EG + 0.211 * FG
[DE] = DE - 0.842 * EG - 0.158 * FG
[DF] = DF - 0.211 * EG + 0.211 * FG
[DG] = DG - EF + 0.263 * EG - 0.263 * FG

12.6. Consider the experiment in Problem 11.8. Suppose that pressure is a noise variable ($\sigma_z^2 = 1$ in coded units). Fit the response model for the viscosity response. Find a set of conditions that result in viscosity as close as possible to 600 and that minimize the variability transmitted from the noise variable pressure.

Design Expert Output

Response: Viscosity

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	85467.33	6	14244.56	12.12	0.0012	significant
A	*703.12*	*1*	*703.12*	*0.60*	*0.4615*	
B	*6105.12*	*1*	*6105.12*	*5.19*	*0.0522*	
C	*5408.00*	*1*	*5408.00*	*4.60*	*0.0643*	
A2	*21736.93*	*1*	*21736.93*	*18.49*	*0.0026*	
C2	*5153.80*	*1*	*5153.80*	*4.38*	*0.0696*	
AC	*47742.25*	*1*	*47742.25*	*40.61*	*0.0002*	
Residual	9404.00	8	1175.50			
Lack of Fit	*7922.00*	*6*	*1320.33*	*1.78*	*0.4022*	*not significant*
Pure Error	*1482.00*	*2*	*741.00*			
Cor Total	94871.33	14				

The Model F-value of 12.12 implies the model is significant. There is only a 0.12% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	34.29	R-Squared	0.9009
Mean	575.33	Adj R-Squared	0.8265
C.V.	5.96	Pred R-Squared	0.6279
PRESS	35301.77	Adeq Precision	11.731

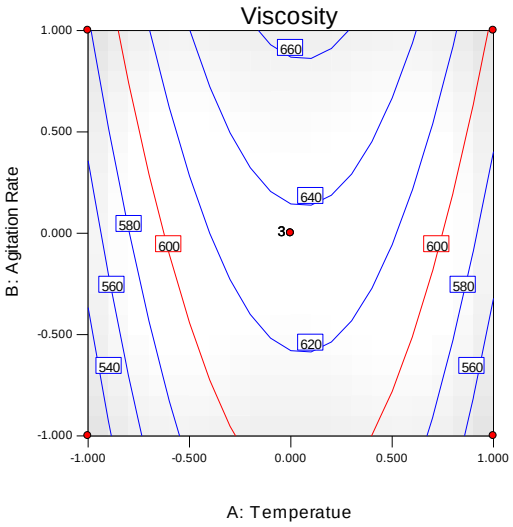
Final Equation in Terms of Coded Factors:

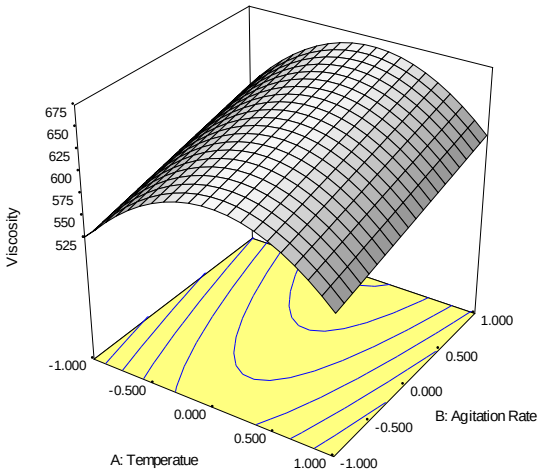
Viscosity =
+636.00
+9.37 * A
+27.62 * B
-26.00 * C
-76.50 * A2
-37.25 * C2
+109.25 * A * C

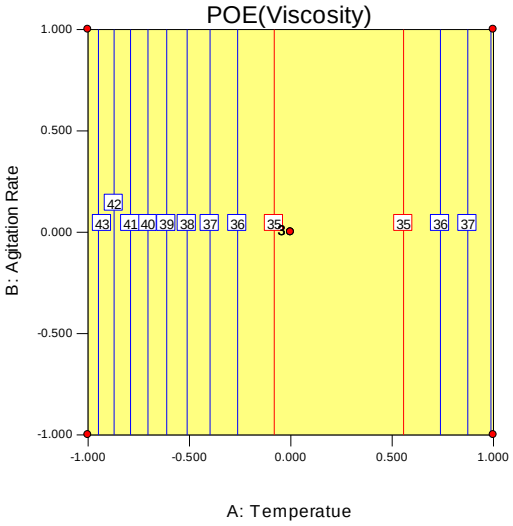
From the final equation shown in the above analysis, the mean model is shown below.

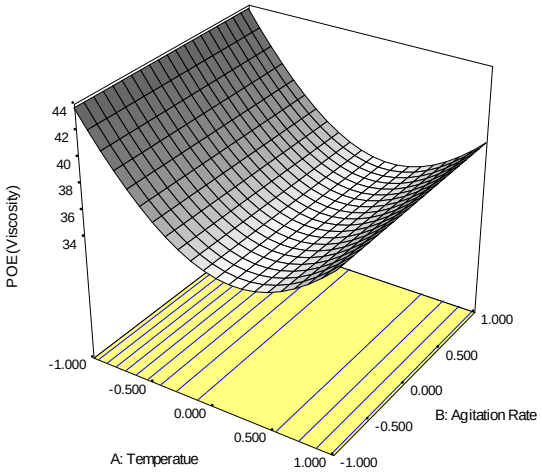
$$E_z\left[y\left(\mathbf{x}, z_1\right)\right] = 636.00 + 9.37x_1 + 27.62x_2 - 26.00x_3 - 76.50x_1^2 - 37.25x_3^2 + 109.25x_1x_3$$

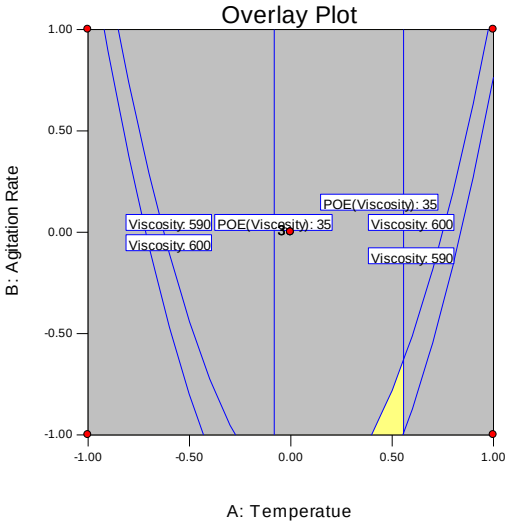
The corresponding contour and 3-D plots for this model are shown below followed by the POE contour and 3-D plots. Finally, the stacked contour plot is presented identifying a region with viscosity between 590 and 610 while minimizing the variability transmitted from the noise variable pressure. These conditions are in the region of factor *A* = 0.5 and factor *B* = -1.


Viscosity
660
640
620
600
600
580
580
560
560
540
3
1.000
0.500
0.000
-0.500
-1.000
B: Agitation Rate
-1.000
-0.500
0.000
0.500
1.000
A: Temperatue


675
650
625
600
575
550
525
Viscosity
-1.000
-0.500
0.000
0.500
1.000
A: Temperatue
1.000
0.500
0.000
-0.500
-1.000
B: Agitation Rate


POE(Viscosity)
42
43
41
40
39
38
37
36
35
35
36
37
3
1.000
0.500
0.000
-0.500
-1.000
B: Agitation Rate
-1.000
-0.500
0.000
0.500
1.000
A: Temperatue


44
42
40
38
36
34
POE(Viscosity)
-1.000
-0.500
0.000
0.500
1.000
A: Temperatue
1.000
0.500
0.000
-0.500
-1.000
B: Agitation Rate


Overlay Plot
POE(Viscosity): 35
Viscosity: 590
POE(Viscosity): 35
Viscosity: 600
Viscosity: 600
Viscosity: 590
3
1.00
0.50
0.00
-0.50
-1.00
B: Agitation Rate
-1.00
-0.50
0.00
0.50
1.00
A: Temperatue

12.7. A variation of Example 12.1. In example 12.1 (which utilized data from Example 6.2) we found that one of the process variables (B = pressure) was not important. Dropping this variable produced two replicates of a 2^3 design. The data are shown below.

C	D	A(-)	A(+)	$\bar{y}$	s^2
–	–	45, 48	71, 65	57.75	121.19
+	–	68, 80	60, 65	68.25	72.25
–	+	43, 45	100, 104	73.00	1124.67
+	+	75, 70	86, 96	81.75	134.92

Assume that C and D are controllable factors and that A is a noise factor.

(a) Fit a model to the mean response.

The following is the analysis of variance with all terms in the model:

Design Expert Output

Response: Mean
ANOVA for Selected Factorial Model
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F
Model	300.05	3	100.02		
A	*92.64*	*1*	*92.64*		
B	*206.64*	*1*	*206.64*		
AB	*0.77*	*1*	*0.77*		
Pure Error	0.000	0			
Cor Total	300.05	3			

Based on the above analysis, the AB interaction is removed from the model and used as error.

Design Expert Output

Response: Mean
ANOVA for Selected Factorial Model
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	299.28	2	149.64	195.45	0.0505	not significant
A	*92.64*	*1*	*92.64*	*121.00*	*0.0577*	
B	*206.64*	*1*	*206.64*	*269.90*	*0.0387*	
Residual	0.77	1	0.77			
Cor Total	300.05	3				

The Model F-value of 195.45 implies there is a 5.05% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.87	R-Squared	0.9974
Mean	70.19	Adj R-Squared	0.9923
C.V.	1.25	Pred R-Squared	0.9592
PRESS	12.25	Adeq Precision	31.672

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	70.19	1	0.44	64.63	75.75	
A-Concentration	4.81	1	0.44	-0.75	10.37	1.00
B-Stir Rate	7.19	1	0.44	1.63	12.75	1.00

Final Equation in Terms of Coded Factors:

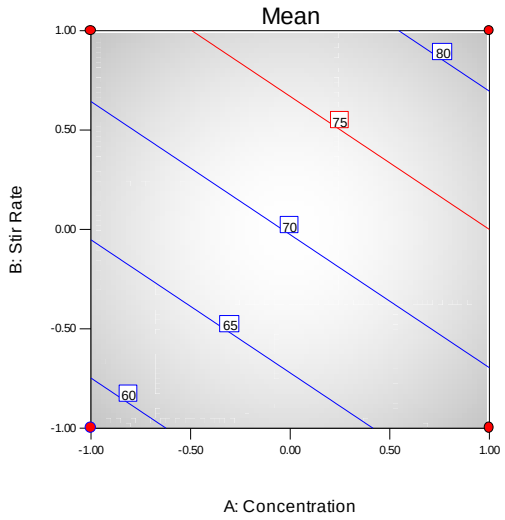
Mean =
+70.19
+4.81 * A

+7.19 * B

Final Equation in Terms of Actual Factors:

Mean =
+70.18750
+4.81250 * Concentration
+7.18750 * Stir Rate

The following is a contour plot of the mean model:



(b) Fit a model to the $\ln(s^2)$ response.

The following is the analysis of variance with all terms in the model:

Design Expert Output

Response: Variance Transform: Natural log Constant: 0

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F
Model	4.42	3	1.47		
A	*1.74*	*1*	*1.74*		
B	*2.03*	*1*	*2.03*		
AB	*0.64*	*1*	*0.64*		
Pure Error	0.000	0			
Cor Total	4.42	3			

Based on the above analysis, the *AB* interaction is removed from the model and applied to the residual error.

Design Expert Output

Response: Variance Transform: Natural log Constant: 0

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3.77	2	1.89	2.94	0.3815	not significant
A	*1.74*	*1*	*1.74*	*2.71*	*0.3477*	
B	*2.03*	*1*	*2.03*	*3.17*	*0.3260*	
Residual	0.64	1	0.64			
Cor Total	4.42	3				

The "Model F-value" of 2.94 implies the model is not significant relative to the noise. There is a 38.15 % chance that a "Model F-value" this large could occur due to noise.

Std. Dev.	0.80	R-Squared	0.8545
Mean	5.25	Adj R-Squared	0.5634
C.V.	15.26	Pred R-Squared	-1.3284
PRESS	10.28	Adeq Precision	3.954

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	5.25	1	0.40	0.16	10.34	
A-Concentration	-0.66	1	0.40	-5.75	4.43	1.00
B-Stir Rate	0.71	1	0.40	-4.38	5.81	1.00

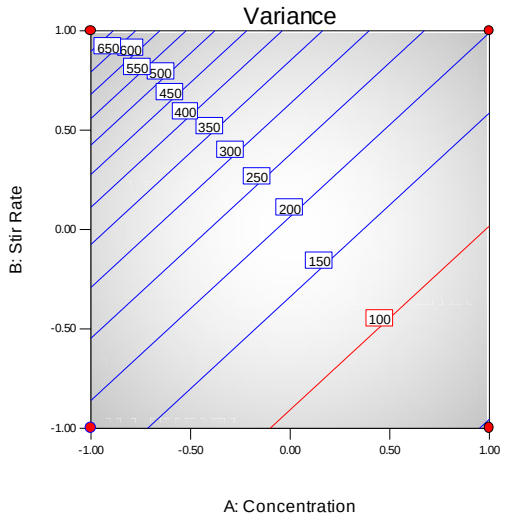
Final Equation in Terms of Coded Factors:

Ln(Variance) =
+5.25
-0.66 * A
+0.71 * B

Final Equation in Terms of Actual Factors:

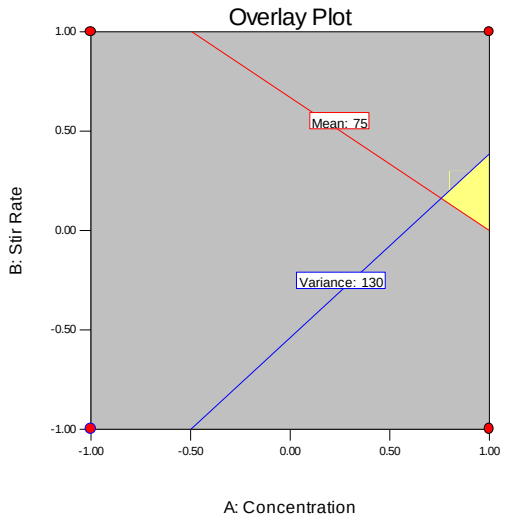
Ln(Variance) =
+5.25185
-0.65945 * Concentration
+0.71311 * Stir Rate

The following is a contour plot of the variance model in the untransformed form:



(c) Find operating conditions that result in the mean filtration rate response exceeding 75 with minimum variance.

The overlay plot shown below identifies the region required by the process:



(d) Compare your results with those from Example 12.1 which used the transmission of error approach. How similar are the two answers.

The results are very similar. Both require the Concentration to be held at the high level while the stirring rate is held near the middle.

12.8S. In an article ("Let's All Beware the Latin Square," *Quality Engineering*, Vol. 1, 1989, pp. 453-465) J.S. Hunter illustrates some of the problems associated with 3^{k-p} fractional factorial designs. Factor *A* is the amount of ethanol added to a standard fuel and factor *B* represents the air/fuel ratio. The response variable is carbon monoxide (CO) emission in g/m^2. The design is shown below.

Design				Observations	
A	*B*	x_1	x_2	*y*	
0	0	-1	-1	66	62
1	0	0	-1	78	81
2	0	1	-1	90	94
0	1	-1	0	72	67
1	1	0	0	80	81
2	1	1	0	75	78
0	2	-1	1	68	66
1	2	0	1	66	69
2	2	1	1	60	58

Notice that we have used the notation system of 0, 1, and 2 to represent the low, medium, and high levels for the factors. We have also used a "geometric notation" of -1, 0, and 1. Each run in the design is replicated twice.

(a) Verify that the second-order model

$$\hat{y} = 78.5 + 4.5x_1 - 7.0x_2 - 4.5x_1^2 - 4.0x_2^2 - 9.0x_1x_2$$

is a reasonable model for this experiment. Sketch the CO concentration contours in the x_1, x_2 space.

In the computer output that follows, the "coded factors" model is in the -1, 0, +1 scale.

Design Expert Output

Response: CO Emis

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	1624.00	5	324.80	50.95	< 0.0001	significant
A	*243.00*	*1*	*243.00*	*38.12*	*< 0.0001*	
B	*588.00*	*1*	*588.00*	*92.24*	*< 0.0001*	
A^2	*81.00*	*1*	*81.00*	*12.71*	*0.0039*	
B^2	*64.00*	*1*	*64.00*	*10.04*	*0.0081*	
AB	*648.00*	*1*	*648.00*	*101.65*	*< 0.0001*	
Residual	76.50	12	6.37			
Lack of Fit	*30.00*	*3*	*10.00*	*1.94*	*0.1944*	*not significant*
Pure Error	*46.50*	*9*	*5.17*			
Cor Total	1700.50	17				

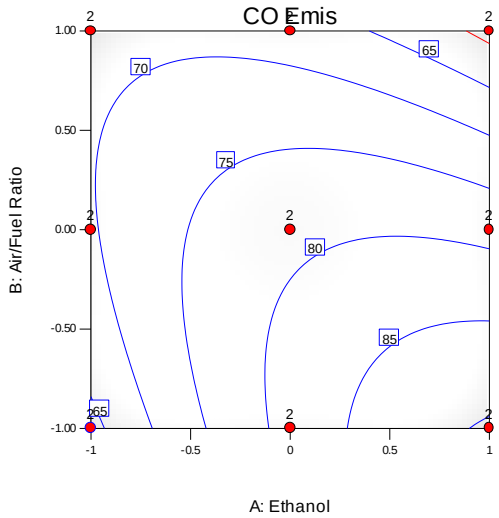
The Model F-value of 50.95 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	2.52	R-Squared	0.9550
Mean	72.83	Adj R-Squared	0.9363
C.V.	3.47	Pred R-Squared	0.9002
PRESS	169.71	Adeq Precision	21.952

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	78.50	1	1.33	75.60	81.40	
A-Ethanol	4.50	1	0.73	2.91	6.09	1.00
B-Air/Fuel Ratio	-7.00	1	0.73	-8.59	-5.41	1.00
A^2	-4.50	1	1.26	-7.25	-1.75	1.00
B^2	-4.00	1	1.26	-6.75	-1.25	1.00
AB	-9.00	1	0.89	-10.94	-7.06	1.00

Final Equation in Terms of Coded Factors:

$$\begin{aligned}\text{CO Emis} = {} & +78.50 \\ & +4.50 * A \\ & -7.00 * B \\ & -4.50 * A^2 \\ & -4.00 * B^2 \\ & -9.00 * A * B\end{aligned}$$



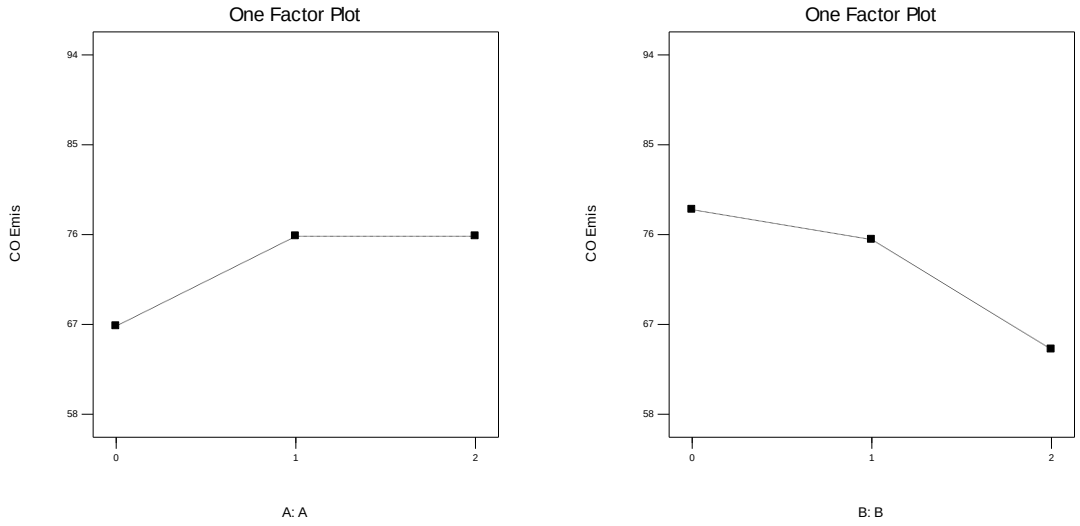
(b) Now suppose that instead of only two factors, we had used *four* factors in a 3^{4-2} fractional factorial design and obtained *exactly* the same data in part (a). The design would be as follows:

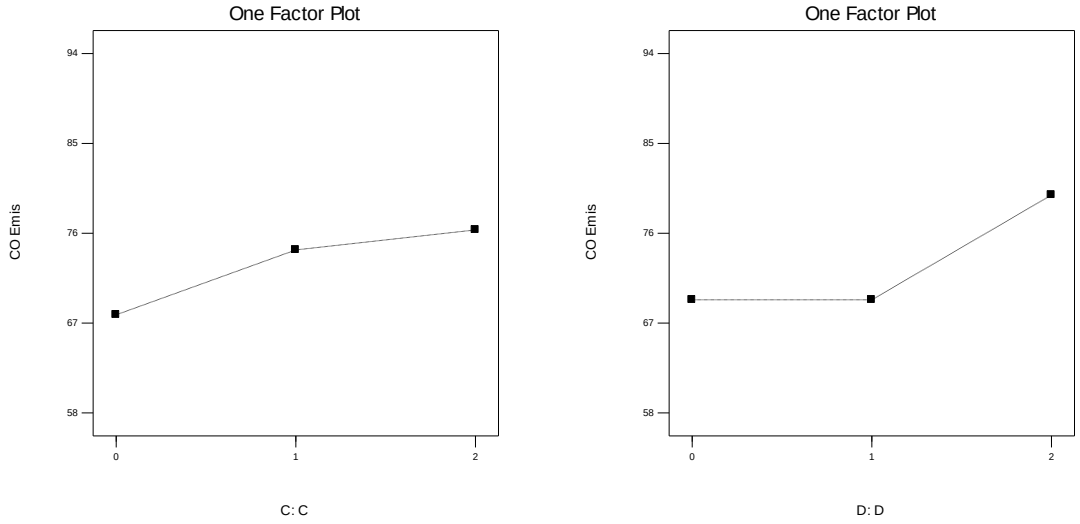
Design								Observations	
A	*B*	*C*	*D*	x_1	x_2	x_3	x_4	*y*	*y*
0	0	0	0	-1	-1	-1	-1	66	62
1	0	1	1	0	-1	0	0	78	81
2	0	2	2	+1	-1	+1	+1	90	94
0	1	2	1	-1	0	+1	0	72	67
1	1	0	2	0	0	-1	+1	80	81
2	1	1	0	+1	0	0	-1	75	78
0	2	1	2	-1	+1	0	+1	68	66
1	2	2	0	0	+1	+1	-1	66	69
2	2	0	1	+1	+1	-1	0	60	58

Calculate the marginal averages of the CO response at each level of four factors *A*, *B*, *C*, and *D*. Construct plots of these marginal averages and interpret the results. Do factors *C* and *D* appear to have strong effects? Do these factors *really* have any effect on CO emission? Why is their appearance effect strong?

The marginal averages are shown below. Both Factors *C* and *D* appear to have an effect on CO emission. This is probably because both *C* and *D* are aliased with components of the interaction involving *A* and *B*, and there is a strong *AB* interaction.

Level	*A*	*B*	*C*	*D*
0	66.83	78.50	67.83	69.33
1	75.83	75.50	74.33	69.33
2	75.83	64.50	76.33	79.83





(c) The design in part (b) allows the model

$$y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_{ii} x_i^2 + \varepsilon$$

to be fitted. Suppose that the *true* model is

$$y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_{ii} x_i^2 + \sum\sum_{i<j} \beta_{ij} x_i x_j + \varepsilon$$

Show that if $\hat{\beta}_j$ represents the least squares estimates of the coefficients in the fitted model, then

$$E\left(\hat{\beta}_0\right) = \beta_0 - \beta_{13} - \beta_{14} - \beta_{34}$$
$$E\left(\hat{\beta}_1\right) = \beta_1 - \left(\beta_{23} + \beta_{24}\right)/2$$
$$E\left(\hat{\beta}_2\right) = \beta_2 - \left(\beta_{13} + \beta_{14} + \beta_{34}\right)/2$$
$$E\left(\hat{\beta}_3\right) = \beta_3 - \left(\beta_{12} + \beta_{24}\right)/2$$
$$E\left(\hat{\beta}_4\right) = \beta_4 - \left(\beta_{12} + \beta_{23}\right)/2$$
$$E\left(\hat{\beta}_{11}\right) = \beta_{11} - \left(\beta_{23} - \beta_{24}\right)/2$$
$$E\left(\hat{\beta}_{22}\right) = \beta_{22} + \left(\beta_{13} + \beta_{14} + \beta_{34}\right)/2$$
$$E\left(\hat{\beta}_{33}\right) = \beta_{33} - \left(\beta_{24} - \beta_{12}\right)/2 + \beta_{14}$$
$$E\left(\hat{\beta}_{44}\right) = \beta_{44} - \left(\beta_{12} - \beta_{23}\right)/2 + \beta_{13}$$

Does this help explain the strong effects for factors *C* and *D* observed graphically in part (b)?

$$\text{Let } \mathbf{X}_1 = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 & \beta_4 & \beta_{11} & \beta_{22} & \beta_{33} & \beta_{44} \\ 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & -1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \text{ and } \mathbf{X}_2 = \begin{bmatrix} \beta_{12} & \beta_{13} & \beta_{14} & \beta_{23} & \beta_{24} & \beta_{34} \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 1 & -1 & -1 & 1 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 1 & -1 & 0 & -1 & 0 & 0 \end{bmatrix}$$

$$\text{Then, } \mathbf{A} = \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{X}_2 = \begin{bmatrix} 0 & -1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1/2 & -1/2 & 0 \\ 0 & -1/2 & -1/2 & 0 & 0 & -1/2 \\ -1/2 & 0 & 0 & 0 & -1/2 & 0 \\ -1/2 & 0 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & -1/2 & 1/2 & 1 \\ 0 & 1/2 & 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 1 & 0 & -1/2 & 0 \\ -1/2 & 1 & 0 & 1/2 & 0 & 0 \end{bmatrix}$$

$$E\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \\ \hat{\beta}_{11} \\ \hat{\beta}_{22} \\ \hat{\beta}_{33} \\ \hat{\beta}_{44} \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_{11} \\ \beta_{22} \\ \beta_{33} \\ \beta_{44} \end{bmatrix} + \begin{bmatrix} 0 & -1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1/2 & -1/2 & 0 \\ 0 & -1/2 & -1/2 & 0 & 0 & -1/2 \\ -1/2 & 0 & 0 & 0 & -1/2 & 0 \\ -1/2 & 0 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & -1/2 & 1/2 & 1 \\ 0 & 1/2 & 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 1 & 0 & -1/2 & 0 \\ -1/2 & 1 & 0 & 1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_{12} \\ \beta_{13} \\ \beta_{14} \\ \beta_{23} \\ \beta_{24} \\ \beta_{34} \end{bmatrix} = \begin{bmatrix} \beta_0 - \beta_{13} - \beta_{14} - \beta_{34} \\ \beta_1 - 1/2\beta_{23} - 1/2\beta_{24} \\ \beta_2 - 1/2\beta_{13} - 1/2\beta_{14} - 1/2\beta_{34} \\ \beta_3 - 1/2\beta_{12} - 1/2\beta_{24} \\ \beta_4 - 1/2\beta_{12} - 1/2\beta_{23} \\ \beta_{11} - 1/2\beta_{23} + 1/2\beta_{24} + \beta_{34} \\ \beta_{22} + 1/2\beta_{13} + 1/2\beta_{14} + 1/2\beta_{34} \\ \beta_{33} + 1/2\beta_{12} + \beta_{14} - 1/2\beta_{24} \\ \beta_{44} - 1/2\beta_{12} + \beta_{13} + 1/2\beta_{23} \end{bmatrix}$$

12.9. An experiment has been run in a process that applies a coating material to a wafer. Each run in the experiment produced a wafer, and the coating thickness was measured several times at different locations on the wafer. Then the mean y_1, and standard deviation y_2 of the thickness measurement was obtained. The data [adapted from Box and Draper (1987)] are shown in the table below.

Run	Speed	Pressure	Distance	Mean (y_1)	Std Dev (y_2)
1	-1	-1	-1	24.0	12.5
2	0	-1	-1	120.3	8.4
3	1	-1	-1	213.7	42.8
4	-1	0	-1	86.0	3.5
5	0	0	-1	136.6	80.4
6	1	0	-1	340.7	16.2
7	-1	1	-1	112.3	27.6

8	0	1	-1	256.3	4.6
9	1	1	-1	271.7	23.6
10	-1	-1	0	81.0	0.0
11	0	-1	0	101.7	17.7
12	1	-1	0	357.0	32.9
13	-1	0	0	171.3	15.0
14	0	0	0	372.0	0.0
15	1	0	0	501.7	92.5
16	-1	1	0	264.0	63.5
17	0	1	0	427.0	88.6
18	1	1	0	730.7	21.1
19	-1	-1	1	220.7	133.8
20	0	-1	1	239.7	23.5
21	1	-1	1	422.0	18.5
22	-1	0	1	199.0	29.4
23	0	0	1	485.3	44.7
24	1	0	1	673.7	158.2
25	-1	1	1	176.7	55.5
26	0	1	1	501.0	138.9
27	1	1	1	1010.0	142.4

(a) What type of design did the experimenters use? Is this a good choice of design for fitting a quadratic model?

The design is a 3^3. A better choice would be a 2^3 central composite design. The CCD gives more information over the design region with fewer points.

(b) Build models of both responses.

The model for the mean is developed as follows:

Design Expert Output

Response: Mean

ANOVA for Response Surface Reduced Cubic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	1.289E+006	7	1.841E+005	60.45	< 0.0001	significant
A	*5.640E+005*	*1*	*5.640E+005*	*185.16*	*< 0.0001*	
B	*2.155E+005*	*1*	*2.155E+005*	*70.75*	*< 0.0001*	
C	*3.111E+005*	*1*	*3.111E+005*	*102.14*	*< 0.0001*	
AB	*52324.81*	*1*	*52324.81*	*17.18*	*0.0006*	
AC	*68327.52*	*1*	*68327.52*	*22.43*	*0.0001*	
BC	*22794.08*	*1*	*22794.08*	*7.48*	*0.0131*	
ABC	*54830.16*	*1*	*54830.16*	*18.00*	*0.0004*	
Residual	57874.57	19	3046.03			
Cor Total	1.347E+006	26				

The Model F-value of 60.45 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	55.19	R-Squared	0.9570
Mean	314.67	Adj R-Squared	0.9412
C.V.	17.54	Pred R-Squared	0.9056
PRESS	1.271E+005	Adeq Precision	33.333

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	314.67	1	10.62	292.44	336.90	
A-Speed	177.01	1	13.01	149.78	204.24	1.00
B-Pressure	109.42	1	13.01	82.19	136.65	1.00
C-Distance	131.47	1	13.01	104.24	158.70	1.00
AB	66.03	1	15.93	32.69	99.38	1.00

AC	75.46	1	15.93	42.11	108.80	1.00
BC	43.58	1	15.93	10.24	76.93	1.00
ABC	82.79	1	19.51	41.95	123.63	1.00

Final Equation in Terms of Coded Factors:

Mean =
+314.67
+177.01 * A
+109.42 * B
+131.47 * C
+66.03 * A * B
+75.46 * A * C
+43.58 * B * C
+82.79 * A * B * C

Final Equation in Terms of Actual Factors:

Mean =
+314.67037
+177.01111 * Speed
+109.42222 * Pressure
+131.47222 * Distance
+66.03333 * Speed * Pressure
+75.45833 * Speed * Distance
+43.58333 * Pressure * Distance
+82.78750 * Speed * Pressure * Distance

The model for the Std. Dev. response is as follows. A square root transformation was applied to correct problems with the normality assumption.

Design Expert Output

Response: Std. Dev. Transform: Square root Constant: 0

ANOVA for Response Surface Linear Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	116.75	3	38.92	4.34	0.0145	significant
A	*16.52*	*1*	*16.52*	*1.84*	*0.1878*	
B	*26.32*	*1*	*26.32*	*2.94*	*0.1001*	
C	*73.92*	*1*	*73.92*	*8.25*	*0.0086*	
Residual	206.17	23	8.96			
Cor Total	322.92	26				

The Model F-value of 4.34 implies the model is significant. There is only a 1.45% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	2.99	R-Squared	0.3616
Mean	6.00	Adj R-Squared	0.2783
C.V.	49.88	Pred R-Squared	0.1359
PRESS	279.05	Adeq Precision	7.278

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	6.00	1	0.58	4.81	7.19	
A-Speed	0.96	1	0.71	-0.50	2.42	1.00
B-Pressure	1.21	1	0.71	-0.25	2.67	1.00
C-Distance	2.03	1	0.71	0.57	3.49	1.00

Final Equation in Terms of Coded Factors:

Sqrt(Std. Dev.) =
+6.00
+0.96 * A
+1.21 * B
+2.03 * C

Final Equation in Terms of Actual Factors:

Sqrt(Std. Dev.) =
+6.00273
+0.95796 * Speed
+1.20916 * Pressure
+2.02643 * Distance

Because Factor *A* is insignificant, it is removed from the model. The reduced linear model analysis is shown below:

Design Expert Output

Response: Std. Dev. Transform: Square root Constant: 0

ANOVA for Response Surface Reduced Linear Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	100.23	2	50.12	5.40	0.0116	significant
B	*26.32*	*1*	*26.32*	*2.84*	*0.1051*	
C	*73.92*	*1*	*73.92*	*7.97*	*0.0094*	
Residual	222.68	24	9.28			
Cor Total	322.92	26				

The Model F-value of 5.40 implies the model is significant. There is only a 1.16% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	3.05	R-Squared	0.3104
Mean	6.00	Adj R-Squared	0.2529
C.V.	50.74	Pred R-Squared	0.1476
PRESS	275.24	Adeq Precision	6.373

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	6.00	1	0.59	4.79	7.21	
B-Pressure	1.21	1	0.72	-0.27	2.69	1.00
C-Distance	2.03	1	0.72	0.54	3.51	1.00

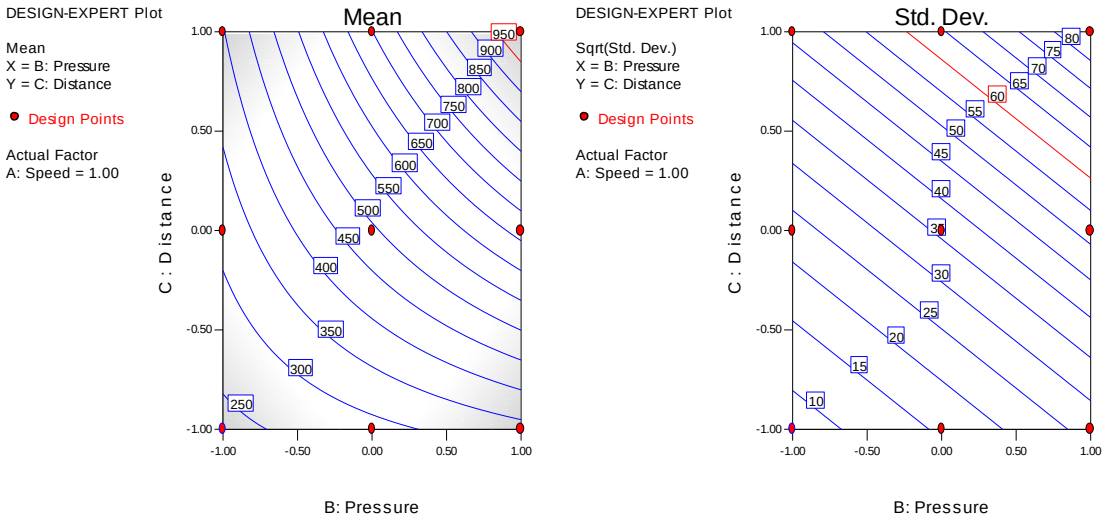
Final Equation in Terms of Coded Factors:

Sqrt(Std. Dev.) =
+6.00
+1.21 * B
+2.03 * C

Final Equation in Terms of Actual Factors:

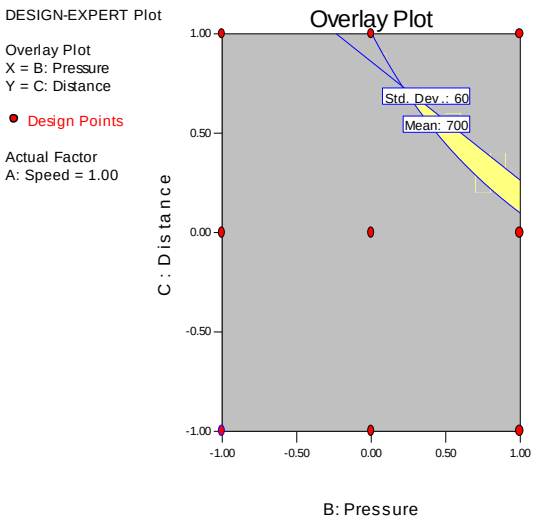
Sqrt(Std. Dev.) =
+6.00273
+1.20916 * Pressure
+2.02643 * Distance

The following contour plots graphically represent the two models.



(c) Find a set of optimum conditions that result in the mean as large as possible with the standard deviation less than 60.

The overlay plot identifies a region that meets the criteria of the mean as large as possible with the standard deviation less than 60. The optimum conditions in coded terms are approximately Speed = 1.0, Pressure = 0.75 and Distance = 0.25.



12.10S. Suppose that there are four controllable variables and two noise variables. It is necessary to estimate the main effects and two-factor interactions of all of the controllable variables, the main effects of the noise variables, and the two-factor interactions between all controllable and noise factors. If all factors are at two levels, what is the minimum number of runs that can be used to estimate all of the model parameters using a combined array design? Use a *D*-optimal algorithm to find a design.

Twenty-one runs are required for the model, with five additional runs for lack of fit, and five as replicates for a total of 31 runs as follows. It should be noted that *Design Expert's D*-optimal algorithm might not create the same design if repeated.

Std	*A*	*B*	*C*	*D*	*E*	*F*
1	+1	+1	-1	+1	+1	+1
2	-1	+1	-1	+1	-1	-1
3	+1	-1	-1	+1	-1	-1
4	+1	+1	-1	-1	-1	+1
5	-1	+1	-1	-1	+1	+1
6	-1	+1	+1	+1	+1	+1
7	+1	+1	-1	-1	+1	-1
8	-1	-1	+1	+1	-1	-1
9	-1	+1	+1	-1	+1	-1
10	-1	+1	+1	-1	-1	+1
11	+1	-1	+1	+1	+1	+1
12	+1	+1	+1	+1	-1	+1
13	+1	-1	-1	-1	+1	+1
14	+1	+1	+1	-1	+1	+1
15	-1	-1	-1	-1	-1	-1
16	+1	+1	+1	+1	+1	-1
17	-1	-1	-1	+1	-1	+1
18	-1	-1	-1	+1	+1	-1
19	-1	-1	+1	-1	+1	+1
20	+1	-1	+1	-1	+1	-1
21	+1	-1	+1	-1	-1	+1
22	+1	+1	+1	-1	-1	-1
23	+1	-1	-1	-1	-1	-1
24	-1	+1	-1	-1	-1	-1
25	+1	+1	-1	-1	-1	-1
26	-1	-1	+1	-1	-1	-1
27	+1	+1	+1	+1	-1	+1
28	-1	-1	-1	+1	-1	+1
29	+1	+1	+1	+1	+1	-1
30	-1	-1	-1	+1	+1	-1
31	-1	+1	-1	-1	+1	+1

12.11S. Suppose that there are four controllable variables and two noise variables. It is necessary to fit a complete quadratic model in the controllable variables, the main effects of the noise variables, and the two-factor interactions between all controllable and noise factors. Set up a combined array design for this by modifying a central composite design.

The following design is a half fraction central composite design with the axial points removed from the noise factors. The total number of runs is forty-eight which includes eight center points.

Std	*A*	*B*	*C*	*D*	*E*	*F*
1	-1	-1	-1	-1	-1	-1
2	+1	-1	-1	-1	-1	+1
3	-1	+1	-1	-1	-1	+1
4	+1	+1	-1	-1	-1	-1
5	-1	-1	+1	-1	-1	+1
6	+1	-1	+1	-1	-1	-1
7	-1	+1	+1	-1	-1	-1
8	+1	+1	+1	-1	-1	+1
9	-1	-1	-1	+1	-1	+1
10	+1	-1	-1	+1	-1	-1
11	-1	+1	-1	+1	-1	-1
12	+1	+1	-1	+1	-1	+1
13	-1	-1	+1	+1	-1	-1
14	+1	-1	+1	+1	-1	+1
15	-1	+1	+1	+1	-1	+1
16	+1	+1	+1	+1	-1	-1
17	-1	-1	-1	-1	+1	+1
18	+1	-1	-1	-1	+1	-1

19	-1	+1	-1	-1	+1	-1
20	+1	+1	-1	-1	+1	+1
21	-1	-1	+1	-1	+1	-1
22	+1	-1	+1	-1	+1	+1
23	-1	+1	+1	-1	+1	+1
24	+1	+1	+1	-1	+1	-1
25	-1	-1	-1	+1	+1	-1
26	+1	-1	-1	+1	+1	+1
27	-1	+1	-1	+1	+1	+1
28	+1	+1	-1	+1	+1	-1
29	-1	-1	+1	+1	+1	+1
30	+1	-1	+1	+1	+1	-1
31	-1	+1	+1	+1	+1	-1
32	+1	+1	+1	+1	+1	+1
33	-2.378	0	0	0	0	0
34	+2.378	0	0	0	0	0
35	0	-2.378	0	0	0	0
36	0	+2.378	0	0	0	0
37	0	0	-2.378	0	0	0
38	0	0	+2.378	0	0	0
39	0	0	0	-2.378	0	0
40	0	0	0	+2.378	0	0
41	0	0	0	0	0	0
42	0	0	0	0	0	0
43	0	0	0	0	0	0
44	0	0	0	0	0	0
45	0	0	0	0	0	0
46	0	0	0	0	0	0
47	0	0	0	0	0	0
48	0	0	0	0	0	0

12.12. Reconsider the situation in Problem 12.11. What is the minimum number of runs that can be used to estimate all of the model parameters using a combined array design? Use a *D*-optimal algorithm to find a reasonable design for this problem.

A minimum of 25 runs is required. The following design is a 36 run D-optimal with six additional runs included for lack of fit and five as replicates. It should be noted that Design Expert's *D*-optimal algorithm might not create the same design if repeated.

Std	*A*	*B*	*C*	*D*	*E*	*F*
1	+1	+1	+1	-1	-1	-1
2	-1	+1	-1	-1	+1	+1
3	-1	+1	+1	+1	-1	+1
4	+1	+1	-1	+1	-1	-1
5	-1	-1	+1	+1	-1	-1
6	-1	+1	-1	-1	-1	-1
7	+1	-1	-1	+1	+1	-1
8	+1	-1	+1	-1	+1	-1
9	+1	+1	-1	+1	+1	+1
10	+1	-1	-1	-1	-1	-1
11	+1	-1	+1	+1	-1	+1
12	-1	+1	-1	+1	+1	-1
13	+1	+1	+1	-1	+1	+1
14	+1	+1	-1	-1	-1	+1
15	+1	+1	+1	+1	+1	-1
16	-1	-1	-1	+1	-1	+1
17	0	-1	-1	-1	+1	-1
18	0	-1	+1	-1	-1	+1
19	0	+1	0	0	0	0
20	0	0	0	-1	0	0
21	0	0	+1	0	0	0
22	-1	+1	+1	-1	+1	-1
23	-1	-1	+1	0	+1	+1
24	+1	+1	-1	-1	+1	-1
25	0	-1	+1	+1	+1	+1
26	+1	-1	-1	-1	+1	+1

27	-1	-1	-1	0	-1	-1
28	+1	-1	0	+1	-1	-1
29	-1	-1	0	+1	+1	-1
30	+1	-1	-1	0	-1	+1
31	-1	0	-1	+1	+1	+1
32	+1	+1	+1	+1	+1	-1
33	+1	-1	+1	-1	+1	-1
34	-1	+1	+1	+1	-1	+1
35	+1	+1	+1	-1	-1	-1
36	+1	+1	-1	+1	+1	+1

12.13. Rework Problem 12.12 using the *I*-criterion to construct the design. Compare this design to the *D*-optimal design in Problem 12.12. Which design would you prefer?

The JMP Output shown below identifies an *I*-optimal design with 25 runs. It should be noted that JMP's *I*-optimal design algorithm might not create the same design if repeated. We would prefer the *I*-optimal over the *D*-optimal because a response surface model is of interest in this problem.

JMP Output

Custom Design
Factors
Add N Factors
1

X1 Continuous
X2 Continuous
X3 Continuous
X4 Continuous
Noise1 Continuous
Noise2 Continuous

Design

Run	X1	X2	X3	X4	Noise1	Noise2
1	1	-1	0	-1	-1	1
2	-1	1	-1	1	0	1
3	0	0	1	1	1	-1
4	1	-1	-1	-1	-1	-1
5	1	1	1	-1	-1	-1
6	1	1	-1	0	-1	1
7	-1	1	1	1	-1	-1
8	1	-1	1	1	0	1
9	-1	1	0	-1	1	1
10	-1	-1	-1	0	-1	1
11	-1	-1	0	-1	0	-1
12	1	0	0	0	1	-1
13	0	0	0	1	-1	1
14	-1	-1	0	1	1	1
15	0	-1	1	0	-1	-1
16	-1	1	1	-1	1	-1
17	1	1	-1	1	1	-1
18	-1	0	1	-1	-1	0
19	0	-1	1	-1	1	1
20	0	1	0	-1	-1	-1
21	1	-1	-1	1	1	1
22	1	1	-1	-1	1	1
23	0	0	-1	0	1	-1
24	-1	-1	-1	1	-1	-1
25	0	1	1	0	1	1

12.14. Rework Problem 12.10S using the *I*-criterion. Compare this to the *D*-optimal design in Problem 12.12. Which design would you prefer?

The JMP Output shown below identifies an *I*-optimal design with 21 runs. It should be noted that JMP's *I*-optimal design algorithm might not create the same design if repeated. Because the problem requires the experiment be run in at only two levels for each variable, and *I*-optimal algorithm in JMP generates center points for continuous variables, categorical variables were used. We would prefer the *D*-optimal over the *I*-optimal assuming that the interest is to identify the important factors more so than to fit a response surface model. Also, the *D*-optimal algorithm in Design Expert will generate a design for this model with continuous variables and still maintain the requirement for running the experiment at only two levels for each variable.

JMP Output

Custom Design
Factors
Add N Factors
1

X1 Continuous
X2 Continuous
X3 Continuous
X4 Continuous
Noise1 Continuous
Noise2 Continuous

Design

Run	X1	X2	X3	X4	Noise1	Noise2
1	1	1	1	1	-1	-1
2	-1	1	-1	-1	1	-1
3	-1	1	-1	1	1	-1
4	-1	1	-1	-1	-1	-1
5	-1	-1	1	1	-1	-1
6	1	-1	-1	1	-1	-1
7	1	1	1	-1	1	-1
8	-1	1	1	1	-1	-1
9	-1	-1	1	-1	-1	1
10	-1	-1	-1	1	-1	-1
11	1	-1	-1	1	-1	-1
12	1	-1	1	-1	-1	-1
13	1	1	-1	-1	1	1
14	1	1	1	1	-1	1
15	-1	1	1	-1	1	1
16	-1	-1	1	-1	-1	1
17	-1	-1	1	1	-1	1
18	-1	1	1	1	1	-1
19	-1	-1	-1	-1	1	1
20	1	1	-1	-1	-1	1
21	-1	-1	-1	1	-1	1
22	1	1	-1	-1	1	1

12.15. An experiment was run in a wave soldering process. There are five controllable variables and three noise variables. The response variable is the number of solder defects per million opportunities. The experimental design employed was the crossed array shown below.

					Outer Array				
					F	-1	1	1	-1
Inner Array					*G*	-1	1	-1	1
A	*B*	*C*	*D*	*E*	*H*	-1	-1	1	1
1	1	1	-1	-1		194	197	193	275
1	1	-1	1	1		136	136	132	136
1	-1	1	-1	1		185	261	264	264
1	-1	-1	1	-1		47	125	127	42
-1	1	1	1	-1		295	216	204	293
-1	1	-1	-1	1		234	159	231	157
-1	-1	1	1	1		328	326	247	322
-1	-1	-1	-1	-1		186	187	105	104

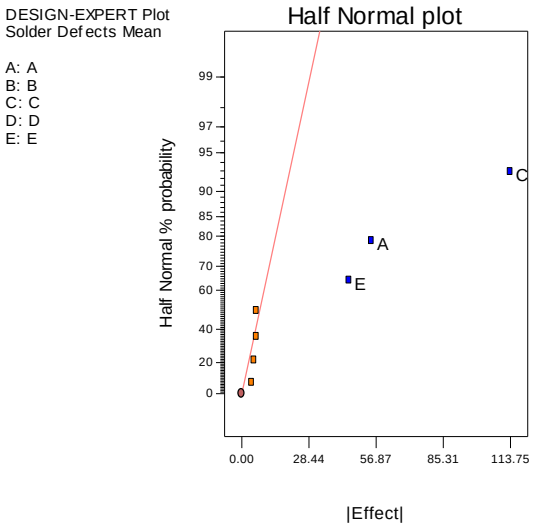
(a) What types of designs were used for the inner and outer arrays?

The inner array is a 2^{5-2} fractional factorial design with a defining relation of $I = -ACD = -BCE = ABDE$. The outer array is a 2^{3-1} fractional factorial design with a defining relation of $I = -FGH$.

(b) Develop models for the mean and variance of solder defects. What set of operating conditions would you recommend?

A	*B*	*C*	*D*	*E*	$\bar{y}$	s^2
1	1	1	-1	-1	214.75	1616.25
1	1	-1	1	1	135.00	4.00
1	-1	1	-1	1	243.50	1523.00
1	-1	-1	1	-1	85.25	2218.92
-1	1	1	1	-1	252.00	2376.67
-1	1	-1	-1	1	195.25	1852.25
-1	-1	1	1	1	305.75	1540.25
-1	-1	-1	-1	-1	145.50	2241.67

The following analysis identifies factors A, C, and E as being significant for the solder defects mean model.



Design Expert Output

Response: Solder Defects Mean
ANOVA for Selected Factorial Model
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	36068.63	3	12022.88	194.31	< 0.0001	significant
A	*6050.00*	*1*	*6050.00*	*97.78*	*0.0006*	
C	*25878.13*	*1*	*25878.13*	*418.23*	*< 0.0001*	
E	*4140.50*	*1*	*4140.50*	*66.92*	*0.0012*	
Residual	247.50	4	61.88			
Cor Total	36316.13	7				

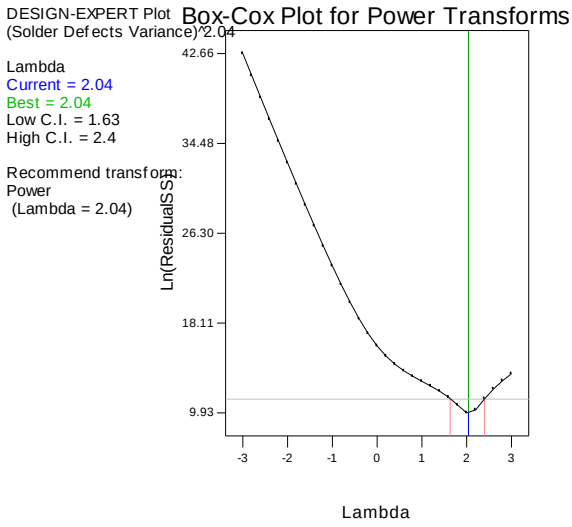
The "Model F-value" of 194.31implies the model is not significant relative to the noise. There is a 0.01 % chance that a "Model F-value" this large could occur due to noise.

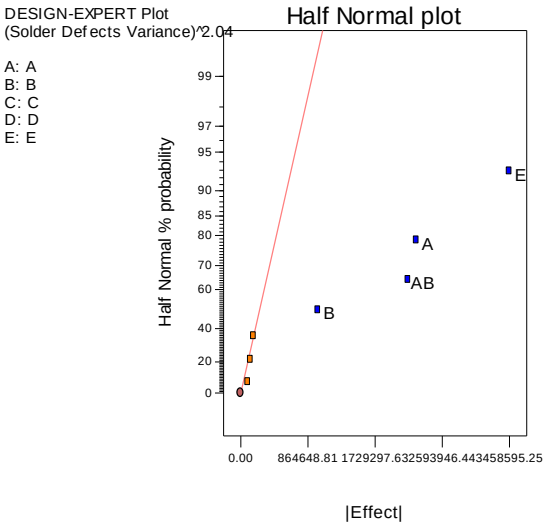
Std. Dev.	7.87	R-Squared	0.9932
Mean	197.13	Adj R-Squared	0.9881
C.V.	3.99	Pred R-Squared	0.9727
PRESS	990.00	Adeq Precision	38.519

Final Equation in Terms of Coded Factors:

Solder Defects Mean =
+197.13
-27.50 * A
+56.88 * C
+22.75 * E

Although the natural log transformation is often utilized for variance response, a power transformation actually performed better for this problem per the Box-Cox plot below. The analysis for the solder defect variance follows.





Design Expert Output

Response: Solder Defects Variance Transform: Power Lambda: 2.04 Constant: 0

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	4.542E+013	4	1.136E+013	325.30	0.0003	significant
A	*1.023E+013*	*1*	*1.023E+013*	*293.08*	*0.0004*	
B	*1.979E+012*	*1*	*1.979E+012*	*56.70*	*0.0049*	
E	*2.392E+013*	*1*	*2.392E+013*	*685.33*	*0.0001*	
AB	*9.289E+012*	*1*	*9.289E+012*	*266.11*	*0.0005*	
Residual	1.047E+011	3	3.491E+010			
Cor Total	4.553E+013	7				

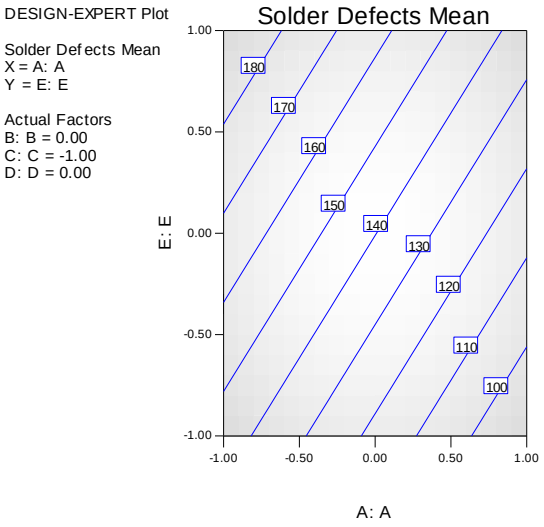
The "Model F-value" of 325.30 implies the model is not significant relative to the noise. There is a 0.03 % chance that a "Model F-value" this large could occur due to noise.

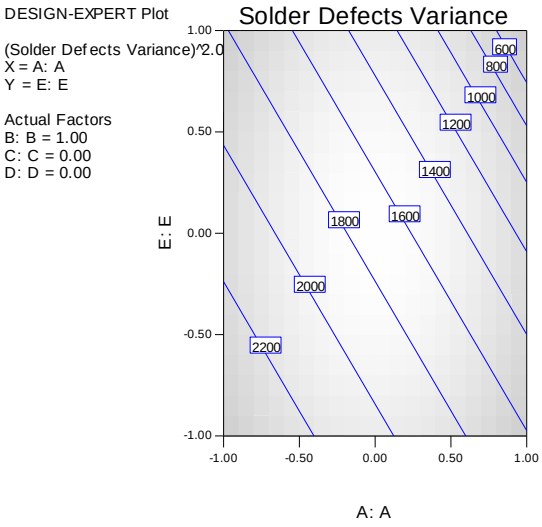
Std. Dev.	1.868E+005	R-Squared	0.9977
Mean	4.461E+006	Adj R-Squared	0.9946
C.V.	4.19	Pred R-Squared	0.9836
PRESS	7.447E+011	Adeq Precision	53.318

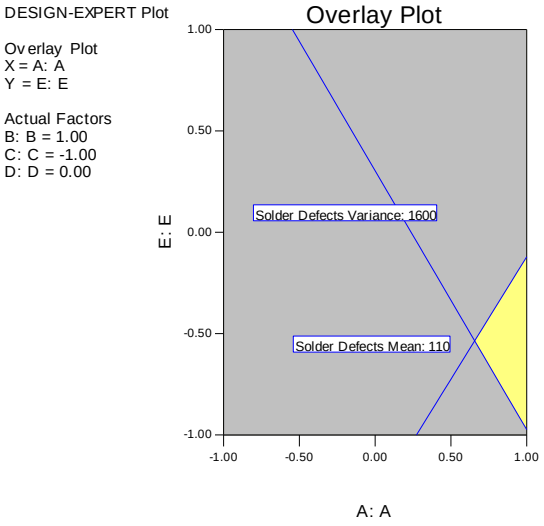
Final Equation in Terms of Coded Factors:

(Solder Defects Variance)2.04 =
+4.461E+006
-1.131E+006 * A
-4.974E+005 * B
-1.729E+006 * E
-1.078E+006 * A * B

The contour plots of the mean and variance models are shown below along with the overlay plot. Assuming that we wish to minimize both solder defects mean and variance, a solution is shown in the overlay plot with factors $A = +1$, $B = +1$, $C = -1$, $D = 0$, and E near -1.







12.16. Reconsider the wave soldering experiment in Problem 12.15. Find a combined array design for this experiment that requires fewer runs.

The following experiment is a 2^{8-4}, resolution IV design with the defining relation $I = BCDE = ACDF = ABCG = ABDH$. Only 16 runs are required.

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*
-1	-1	-1	-1	-1	-1	-1	-1
+1	-1	-1	-1	-1	+1	+1	+1
-1	+1	-1	-1	+1	-1	+1	+1
+1	+1	-1	-1	+1	+1	-1	-1
-1	-1	+1	-1	+1	+1	+1	-1
+1	-1	+1	-1	+1	-1	-1	+1
-1	+1	+1	-1	-1	+1	-1	+1
+1	+1	+1	-1	-1	-1	+1	-1
-1	-1	-1	+1	+1	+1	-1	+1
+1	-1	-1	+1	+1	-1	+1	-1

-1	+1	-1	+1	-1	+1	+1	-1
+1	+1	-1	+1	-1	-1	-1	+1
-1	-1	+1	+1	-1	-1	+1	+1
+1	-1	+1	+1	-1	+1	-1	-1
-1	+1	+1	+1	+1	-1	-1	-1
+1	+1	+1	+1	+1	+1	+1	+1

12.17. Reconsider the wave soldering experiment in Problem 12.15. Suppose that it was necessary to fit a complete quadratic model in the controllable variables, all main effects of the noise variables, and all controllable variable-noise variable interactions. What design would you recommend?

The following experiment is a small central composite design with five center points; the axial points for the noise factors have been removed. A total of 45 runs are required.

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*
+1	+1	+1	-1	-1	+1	+1	+1
-1	+1	+1	-1	+1	+1	+1	-1
+1	+1	-1	-1	+1	+1	-1	-1
+1	-1	-1	+1	+1	-1	+1	-1
-1	-1	+1	+1	+1	+1	-1	-1
-1	+1	+1	+1	-1	+1	-1	-1
-1	+1	+1	+1	+1	+1	-1	-1
+1	+1	+1	+1	+1	-1	+1	-1
+1	+1	-1	+1	-1	+1	-1	+1
+1	-1	+1	+1	-1	-1	+1	-1
+1	+1	+1	-1	+1	-1	-1	+1
-1	+1	+1	-1	-1	-1	+1	+1
+1	-1	-1	-1	-1	+1	-1	-1
+1	-1	+1	-1	-1	-1	-1	+1
-1	+1	-1	-1	+1	+1	+1	-1
+1	-1	-1	+1	+1	+1	-1	+1
-1	-1	-1	-1	-1	+1	+1	-1
-1	-1	+1	+1	+1	-1	-1	+1
-1	+1	-1	-1	+1	-1	+1	+1
-1	-1	+1	+1	-1	+1	+1	+1
+1	+1	-1	+1	-1	-1	+1	-1
-1	-1	+1	-1	+1	-1	-1	-1
+1	+1	+1	-1	-1	+1	-1	-1
+1	-1	-1	-1	+1	+1	+1	+1
+1	-1	+1	-1	-1	+1	+1	+1
-1	+1	-1	+1	+1	+1	+1	+1
-1	-1	-1	-1	+1	-1	+1	+1
+1	-1	+1	+1	+1	-1	+1	-1
-1	+1	-1	+1	-1	-1	-1	+1
-1	-1	-1	-1	-1	-1	-1	-1
-2.34	0	0	0	0	0	0	0
2.34	0	0	0	0	0	0	0
0	-2.34	0	0	0	0	0	0
0	2.34	0	0	0	0	0	0
0	0	-2.34	0	0	0	0	0
0	0	2.34	0	0	0	0	0
0	0	0	-2.34	0	0	0	0
0	0	0	2.34	0	0	0	0
0	0	0	0	-2.34	0	0	0
0	0	0	0	2.34	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

12.18. Signal-to-noise factors generally confound location and dispersion effects.

True False

12.19. A crossed array design allows estimation of the control-by-noise factor interactions.

True False

12.20. If a 3^{3-1} design is used as the inner array and a 2^3 is used as the outer array, all interactions can be estimated.

True **True**

12.21. Suppose that the response model is

$$\hat{y} = 6 + 4x_1 + 3x_2 + 3z + 2x_1z + 2x_2z$$

The value of the controllable factor x_1 that minimizes the transmitted variance is

(a) **–1**
(b) +1
(c) –0.5
(d) None of the above

12.22. Consider the response model in Problem 12.25. The value of the controllable factor x_2 that minimizes the transmitted variance is

(a) **–1**
(b) +1
(c) –0.5
(d) None of the above

12.23. A log transformation could be useful in directly modeling the variance as a response.

True False

12.24. Computer-generated optimal designs can be a useful approach for designing combined array experiments with both control and noise variables.

True False